Optimization for Deep Learning (OPT4DL)

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Tuning Learning Rates

Motivation



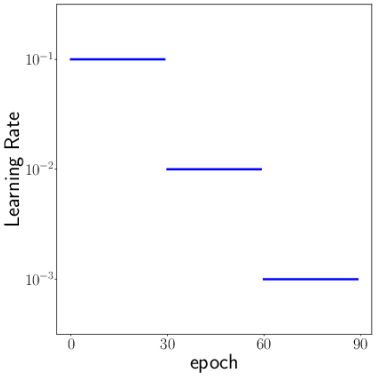
- Recall we showed minibatch SGD satisfies

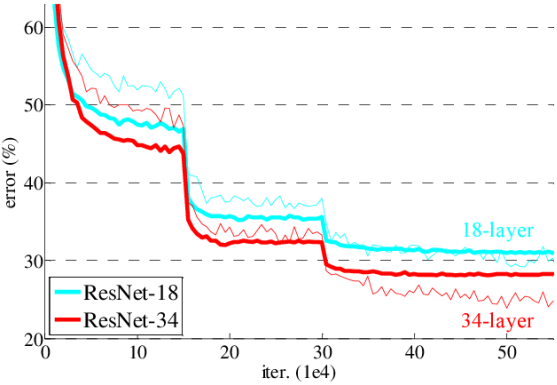
$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_{z_{1:T}}\|\nabla f(w_t)\|^2 \lesssim \frac{\Delta}{T\eta} + \frac{L\eta\sigma^2}{2B} \tag{1}$$

  and we set the $\eta = \min\{\frac{1}{L}, \sqrt{\frac{2\Delta B}{L\sigma^2 T}}\}$ to get the best bound
- A lot of unknown problems. Common practice: set η to be a small constant.
- As we train the model, the first term disappears, but the second term remains.
- Fix: adopt a decreasing schedule of learning rates
  - sublinear decay: $\eta \propto \frac{1}{\sqrt{t}}$
  - exponential decay: $\eta \leftarrow \eta/\alpha$, $\alpha > 1$, after every E epochs
- Interpretation: Early phase of training, we may prefer larger progresses (large η). But, as we approach the solution, we may need a more refined search (small η)
- To find the constants (e.g., the first learning rate), use Hyperparameter search

Exponential Decay

- Frequently used when training ResNets



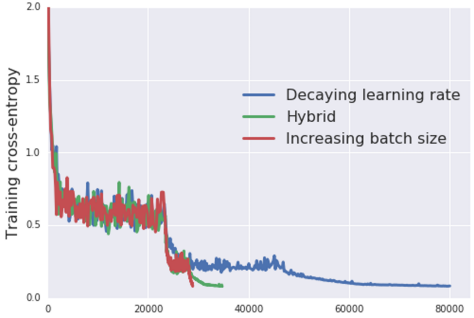


Role of Batch size

- Recall our bound again

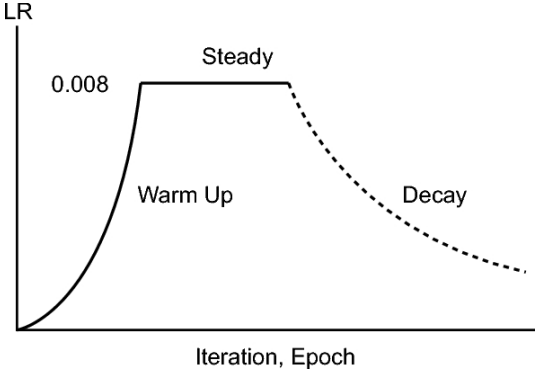
$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_{z_{1:T}}\|\nabla f(w_t)\|^2 \lesssim \frac{\Delta}{T\eta} + \frac{L\eta\sigma^2}{2B} \tag{2}$$

- Conceptually, even if η is constant, we can shrink the second term by increasing the batch size!
- Interpretation: Early phase of training, we may tolerate more noise (smaller B) in our gradient estimation. But, as we approach the solution, we may need a more refined estimation (larger B)



Warm-up

- Gradually increase η to a target and then switch to a decaying schedule
- Can be interpreted as an attempt to bridge the gap between SGD and minibatch SGD



Warm-up (Cont'd)



- Recall, we argued when $\eta = \min\{\frac{1}{L}, \sqrt{\frac{2\Delta B}{L\sigma^2 T}}\}$, in terms of a fixed budget (oracle complexity)

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_{z_{\mathbf{1}:T}}\|\nabla f(w_t)\|^2 \lesssim \frac{B}{C} + \frac{1}{\sqrt{C}} \tag{3}$$

  $B = 1$ leads to a better worst-case bound, i.e., B steps of SGD better than one step of B-SGD
- Warm up could help bridge this gap in the larger gradient regime (early phase of training)
- Assume a small $\eta \propto \frac{B}{\sqrt{C}}$ (tuning constants properly) and recalling $C = B \times T$, from our bound

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_{z_{\mathbf{1}:T}}\|\nabla f(w_t)\|^2 \lesssim \frac{\Delta}{T\eta} + \frac{L\eta\sigma^2}{2B} \tag{4}$$

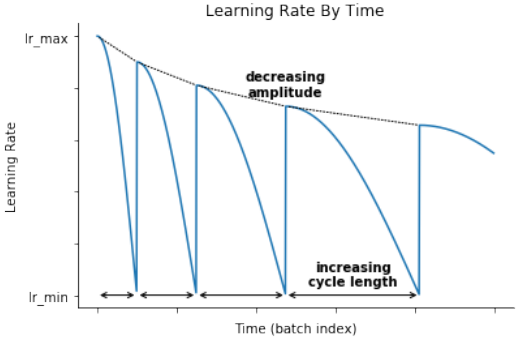
  we can see the resultant worst-case bound becomes $\mathcal{O}(\sqrt{\frac{1}{C}})$, independent of batch size
- In small gradient regime (intermediate and final learning phases), gradient could be small w.r.t. to the noise in gradient estimation, and likely SGD and B-SGD act similarly

Cyclic Learning Rates

- Repeated reset the model with the hope of a better starting point each time

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})[1 + \cos(\frac{t\pi}{T})] \tag{5}$$

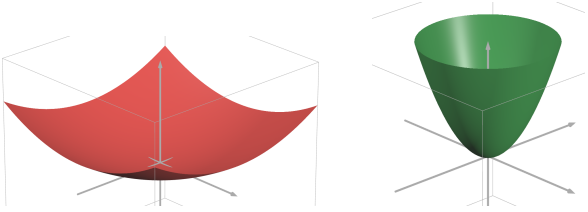
also called cosine-annealing or periodic schedule



Cyclic Schedule as Multi-Step Risk Minimizer

- At any point in the landscape w_t, there are two bad things could happen in the immediate next update w_{t+1}
  - We are at a flat landscape and are using a small η
  - We are at a steep landscape and are using a large η

  First one means slow progress and it's bad, but second one could lead to divergence and that is terrible



- With a constant η, we hedge against the second, by minimizing the one-step risk
- Cyclic Schedule could minimize Multi-Step Risks: what could happen n-th step from now

One-step Risk



- Consider GD and Convex Quadratic losses

$$f(w) = \frac{1}{2} w^\top Q w, \quad LI_d \succeq Q \succeq \mu I_d, \quad w^* = 0, \quad w^+ = (I_d - \eta Q) w \tag{6}$$

- Consider the one-step risk

$$\mathcal{R}(\eta) := \frac{\|w^+ - w^*\|}{\|w - w^*\|} = \frac{\|(I_d - \eta Q) w\|}{\|w\|} = \max_{i=1,\dots,d} |\lambda_i(I_d - \eta Q)| = \max\{1 - \eta\mu, L\eta - 1\} \tag{7}$$

- And the best 1-step η can be found by solving the following linear program (solution where the two line intersect)

$$\eta_1^* = \arg \min_{0 \le \eta \le 1} \max\{1 - \eta\mu, L\eta - 1\} = \frac{2}{\mu + L} \tag{8}$$

  can be calculated directly by analyzing GD (typical in classic convex optimization)

- Now consider a case where we are interested in 2-step risk minimization:

$$w^+ = (I_d - \eta^+ Q)w, \quad w^\# = (I_d - \eta^\# Q)w^+,$$
$$\eta_1^*, \eta_2^* = \arg\min_{\eta^+, \eta^\#} \mathcal{R}(\eta^+, \eta^\#) := \frac{\|w^\# - w^*\|}{\|w - w^*\|} = \max_{i=1,\dots,d} |\lambda_i(I_d - (\eta^+ + \eta^\#)Q + \eta^+\eta^\# Q^2)| \tag{9}$$

A more involved problem but can be solved using Chebyshev polynomials

$$1/\eta_1^* = \frac{\mu + L}{2} + \frac{L - \mu}{2\sqrt{2}}, \quad 1/\eta_2^* = \frac{\mu + L}{2} - \frac{L - \mu}{2\sqrt{2}} \tag{10}$$

- That is, cycle between two learning rates
- Analyze for general training problems still open question...